

## Chapter 4

# Special Probability Distributions

### Learning Objectives

- Understand the assumptions for each of the discrete and continuous probability distributions presented.
- Select an appropriate discrete and continuous probability distribution to calculate probabilities in specific applications.
- Calculate probabilities, determine means and variances for each of the discrete and continuous probability distributions presented.

## 4.0 Special Probability Distributions

### 4.1 Introduction

In general, it is useful to present the probability distribution of a random variable by a model. Probability calculations can then be made convenient by substituting appropriate values into the algebraic model.

In this section, we shall *briefly* define certain special distributions that are widely used in applications of probability and statistics. Distributions that would be considered include both discrete and continuous univariate distributions.

## 4.2 Discrete Probability Distribution

### 4.1.1 Bernoulli Distribution:

A single trial of an experiment may result in one of the two mutually exclusive outcomes such as defective and non-defective, dead or alive, yes or no, male or female, etc. Such a trial is called Bernoulli trial and a sequence of these trials form a Bernoulli process, satisfying the following conditions:

- (i). Each trial results in one of the two mutually exclusive outcomes, *success* and *failure*.
- (ii). The probability of a success,  $p$  remains constant, from trial to trial. The probability of failure is denoted by  $q = 1 - p$ .
- (iii). The trials are independent. That is, the outcome of any particular trial is not affected by the outcome of any other trial.

#### Definition:

A random variable, is said to have a Bernoulli distribution if it assumes the values 0 and 1 for the two outcomes. The probability distribution for the success in the trial is defined by

$$P(x) = p^x(1 - p)^{1-x}, \quad x = 0 \text{ or } 1 \text{ and } 0 < p < 1$$

where the mean and variance of the distribution are as follows:

$$\mu = E(X) = p;$$

$$\sigma^2 = \text{Var}(X) = p(1 - p)$$

An important distribution arising from counting the number of successes in a fixed number of independent Bernoulli trials is the Binomial distribution.

### Example 35

An urn contains 5 red and 15 green balls. Draw one ball at random from the urn. Let  $X=1$  if the ball drawn is red, and  $X=0$  if a green ball is drawn. Obtain

- (i) the p.d.f. of  $X$ ,
- (ii) mean of  $X$  and
- (iii) variance of  $X$ .

### Solution

The p.d.f. of a Bernoulli distribution is  $f(x) = p^x q^{1-x}$ ,  $x = 0, 1$  where

$$p = \frac{5}{20}$$

and

$$q = \frac{15}{20}$$

$$f(x) = \left(\frac{5}{20}\right)^x \left(\frac{15}{20}\right)^{1-x}, \quad x = 0, 1$$

Mean of

$$\bar{X} = E(X) = \sum_{x=0}^1 x \left(\frac{5}{20}\right)^x \left(\frac{15}{20}\right)^{1-x} = (0) \left(\frac{5}{20}\right)^0 \left(\frac{15}{20}\right)^1 + (1) \left(\frac{5}{20}\right)^1 \left(\frac{15}{20}\right)^0$$

$$= \left(\frac{5}{20}\right)$$

Variance of  $X$  =

$$V(X) = \sum_0 x^2 f(x) - [E(X)]^2 = (1) \left(\frac{5}{20}\right)^1 \left(\frac{15}{20}\right)^0 - \left(\frac{5}{20}\right)^2$$

$$= \left(\frac{5}{20}\right) - \left(\frac{5}{20}\right)^2 = \left(\frac{3}{16}\right)$$

## 4.1.2 The Binomial Distribution

The binomial distribution is a discrete probability distribution, where the experiment is repeated  $n$  times under identical conditions and each of the  $n$  trials is independent of each other which results in one of the two outcomes.

Thus, in the event of independent trials (often called Bernoulli trials) let  $p$  be the probability that an event will happen (success) and  $q = 1 - p$  the probability that the event will fail in any single trial. Such experiments are called Binomial experiments and the probability that the event will happen exactly  $x$  times in  $n$  trials is given by the probability function:

$$f(x) = \Pr(X = x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

where the random variable denotes the number of success in  $n$  trials and  $x = 0, 1, 2, 3, \dots, n$ .

The shape of the distribution depends on the two parameters  $n$  and  $p$ .

- (i) when  $p < 0.5$  and  $n$  is small, the distribution will be skewed to the right.
- (ii) when  $p > 0.5$  and  $n$  is small, the distribution will be skewed to the left
- (iii) when  $p = 0.5$  the distribution will be symmetric.
- (iv) In all cases, as  $n$  gets larger the distribution gets closer to being a symmetric, bell-shaped distribution.

### Properties

Mean  $\mu = np$

Variance  $\sigma^2 = npq$

Standard Deviation  $\sigma = \sqrt{npq}$

#### Example 36

If 20% of the bolts produced by a machine are bad. Determine the probability that out of 4 bolts chosen at random.

- (i) one is defective
- (ii) none is defective
- (iii) at most 2 bolts will be defective.

#### Solution

$$n = 4, p = 0.2, q = 0.8$$

$$P[X = 1] = f(1) = \binom{4}{1} 0.2^1 0.8^3 = 0.4096$$

$$P[X = 0] = f(0) = \binom{4}{0} 0.2^0 0.8^4 = 0.4096$$

$$P[X \leq 2] = P[X = 0, 1, 2] = P[X = 0] + P[X = 1] + P[X = 2]$$

$$0.4096 + 0.4096 + 0.1536 = 0.9728$$

$$\text{or } = 1 - P[X > 2] = 1 - P(X = 3) - P(X = 4)$$

$$\text{or } 1 - P[X \geq 3] =$$

$$1 - \binom{4}{3} 0.2^3 0.8^1 - \binom{4}{4} 0.2^4 0.8^0$$

$$= 1 - 0.0256 - 0.0016$$

$$= 0.9728$$

#### Example 37

(a) Suppose that it is known that 30% of a certain population is immune to some disease. If a random sample of 10 is selected from this population. What is the probability that it will contain exactly 4 immune persons?

$$n = 10, p = 0.3, x = 4$$

$$f(4) = \binom{10}{4} (0.3)^4 (0.7)^6$$

$$= 0.2$$

(b) In a certain population 10% of the population is color-blind. If a random sample of 25 people is drawn from this population (use table). Find the probability that

i.  $P(X \geq 5) = 1 - P(X < 5) = 0.0980$

ii.  $P(X \leq 4) = 0.902$  or  $1 - P(X \geq 5) = 1 - 0.0980 = 0.902$

iii.  $P(6 \leq X \leq 10) = p(6) + p(7) + p(8) + \dots + p(10) = 0.0333$

$$= 0.0334 \text{ i.e. } P(X \geq 6)$$

### Example 38

From the experiment “toss four coins and count the number of tails” what is the variance of X?

$n = 4$ ,  $p =$ ,  $q =$

$V(X) = npq = 4 \times \dots = 1$

### Example 39

Roll a fair 6 – sided die 20 times and count the number of times that 6 shows up. What is the standard development of your random variable?

$n = 20$ ,  $p =$

$$\frac{1}{6}$$

$q =$

$$\frac{5}{6}$$

$$V(X) = npq$$

$=$

$$20 \times \frac{1}{6} \times \frac{5}{6} = \frac{100}{36}$$

$\sigma =$

$$\sqrt{V(X)} = \sqrt{\frac{100}{36}} = \frac{10}{6}$$

### Example 40

The following data are the number of seeds germinating out of 10 on damp filter paper for 80 sets of seeds. Fit a binomial distribution to these data.

| x | 0 | 1  | 2  | 3  | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Total |
|---|---|----|----|----|---|---|---|---|---|---|----|-------|
| f | 6 | 20 | 28 | 12 | 8 | 6 | 0 | 0 | 0 | 0 | 0  | 80    |

### Solution

Here  $n = 10$ ,  $N = 80$  and  $\sum f_i = 80$

Arithmetic mean =

$$\frac{\sum f_i x_i}{\sum f_i} = \frac{174}{80}$$

$np =$

$$\frac{174}{80} \therefore p = \frac{174}{80n} = \frac{174}{800} = 0.2175, q = 1 - p = 0.7825$$

Hence the binomial distribution to be fitted is  $b(x; 10, 0.2175)$ . These are approximately,

| x | 0    | 1     | 2     | 3     | 4    | 5    | 6    | 7   | 8    | 9    | 10   | Total |
|---|------|-------|-------|-------|------|------|------|-----|------|------|------|-------|
| f | 6.89 | 19.14 | 23.94 | 17.74 | 8.63 | 2.88 | 0.67 | 0.1 | 0.01 | 0.00 | 0.00 | 80    |

#### 4.1.3 Negative Binomial Distribution (Pascal's Distribution)

Let us consider an experiment in which the properties are the same as those listed for a binomial experiment with the exception that the trials will be repeated until a fixed number of successes occur. Therefore, instead of finding the probability of  $x$  successes in  $n$  trials, where  $n$  is fixed, we are now interested in the probability that the...  $k^{\text{th}}$  success occurs on the  $x^{\text{th}}$  trial. Experiments of this kind are called 'negative binomial experiments'. (Walpole and Myres, 1993). The number  $X$  of trials to produce  $k$  successes in a negative binomial experiment is called a "negative binomial random variable" and its probability distribution is called the "negative binomial distribution".

Since its probabilities depend on the number of successes desired and the probability of success on a given trial, we shall denote them by the symbol  $b^*(x; k, p)$ . For the general formula  $b^*(x; k, p)$ , consider the probability of a success on the trial preceded by  $k-1$  successes and  $x-k$  failures in some specified order. The probability for the specified order ending in success is  $p^{k-1}q^{x-k}p = p^kq^{x-k}$ . The total number of sample points in the experiment ending in success, after the occurrence of  $k-1$  successes and  $x-k$  failures in any order is equal to the number of partitions of  $x-1$  trials into two groups with  $k-1$  successes corresponding to one group and  $x-k$  failures corresponding to the other group. This number is given by the term  $\binom{x-1}{k-1}$ , each mutually exclusive and occurring with equal probability  $p^kq^{x-k}$ . We obtain the general formula by multiplying  $p^kq^{x-k}$  by  $\binom{x-1}{k-1}$ .

In other words:

$$b^*(x; k, p) = \binom{x-1}{k-1} p^{k-1} q^{x-k} p = \binom{x-1}{k-1} p^k q^{x-k} \quad x = k, k+1, \dots$$

$p$  = probability of success

$q = (1 - p)$  = probability of failure

$x$  = total number of trials on which the  $k^{\text{th}}$  success occurs.

**Areas of application** of negative binomial distribution include many biological situations such as death of insects, number of insect bites per fruit (e.g. mango).

##### Example 41

Consider an exploration company that is determined to discover two new fields in a virgin basin it is prospecting, and will drill as many holes as required to achieve its goal. We can investigate the probability that it will require 2, 3, 4, ...,  $n$  exploratory holes before two discoveries are made. The same conditions that govern the binomial distribution may be assumed, except that the number of trials is not fixed.

##### Solution

$x$  dry holes will be drilled before  $r$  discoveries are made.

$$\begin{aligned} P &= \binom{r+x-1}{x} (1-p)^x p^r \\ &= \frac{(r+x-1)!}{(r-1)! x!} (1-p)^x p^r \end{aligned}$$

If the regional success ratio is assumed to be 10% then the probability that a two-hole program

will meet the company's goal of two discoveries will be:

$$\begin{aligned} P &= \frac{(2+0-1)!}{(2-1)!0!}(1-0.1)^0(0.1)^2 \\ &= \frac{1!}{1!0!} \times 0.9^0 \times 0.1^2 \\ &= 1 \times 1 \times 0.01 = 0.01 \end{aligned}$$

The probability that five holes will be required to achieve two successes is:

$$\begin{aligned} p &= \frac{(2+3-1)!}{(2-1)!3!}(1-0.1)^3(0.1)^2 \\ &= \frac{24}{1 \times 6} \times 0.729 \times 0.01 = 0.029 \end{aligned}$$

or

$$p \binom{x-1}{k-1} p^k q^{x-k} = \binom{4}{1} (0.1)^2 (0.9)^3 = \frac{4!}{(4-1)!1!} 0.01 \times 0.729 = 0.029$$

#### Example 42

Find the probability that a person tossing three coins will get either all heads or all tails for the second time in the fifth toss?

*Solution*

$$\begin{aligned} x &= 5, \quad k = 2, \quad p = \left( \frac{1}{8} + \frac{1}{8} \right) = \frac{1}{4} \\ b^*(5; 2, \frac{1}{4}) &= \binom{4}{1} \left( \frac{1}{4} \right)^2 \left( \frac{3}{4} \right)^3 = \frac{27}{256} \end{aligned}$$

The negative binomial distribution derives its name from the fact that each term in the expansion of  $p^k(1-p)^{x-k}$  corresponds to the values of  $b^*(x; k, p)$  for  $x = k, k+1, k+2, \dots$

#### 4.1.4 Geometric Distribution

The geometric distribution is a special case of the negative binomial distribution for which  $k = 1$ . This is the probability distribution for the number of trials required for a single success.

Thus:

$$B^*(x; 1, p) = p q^{x-1}$$

The geometric distribution is denoted by:

$$g(x, p) = p q^{x-1}$$

#### Example 43

In a certain theodolite manufacturing process, it is known that on the average, 1 in every 100 is defective. What is the probability that the fifth item inspected is the first defective theodolite found?

**Solution**

Using the geometric distribution with  $x = 5$ ;  $p = 0.01$ , we have:

$$g(5, 0.01) = p(1 - p)^{5-1} = 0.01 \times 0.99^4 = 0.0096$$

For geometric distribution:

$$\mu = \frac{1}{p} = 100; \quad \sigma^2 = \frac{1-p}{p^2} = 9900$$

**Example 44**

Consider the case in **Example 4.4**, what is the probability that the first discovery will be the fifth exploration hole?

**Solution**

$$n = 5; p = 0.1$$

$$\therefore g(5, 0.1) = p(1 - 0.1)^{5-1} = 0.1 \times 0.9^4 = 0.0656$$

**4.1.5 Poisson Distribution****Introduction:**

Experiments yielding numerical values of a random variable ( $x$ ), the number of successes occurring during a given time interval or in a specified region, are often called poisson experiments. The given time interval may be of any length, such as a minute, a day, a week, a month or even a year. Hence, a poisson experiment might generate observations for the random variable representing the number of telephone calls per hour received by an office, the number of days school is closed due to snow during the winter, or the number of postponed games due to rain during a basketball season. The specified region could be a line segment, an area, a volume or perhaps a material. In this case, might represent the number of field mice per acre, the number of bacteria in a given culture, or the number of typing errors per page.

**The Poisson process:**

A poisson experiment is derived from the Poisson process and possesses the following properties:

- The number of successes occurring in one time interval or specified region are independent of those occurring in any other disjoint time interval or region of space.
- The probability of a single success occurring during a very short time interval or in a small region is proportional to the length of the time interval or the size of the region and does not depend on the number of successes occurring outside this time interval or region.
- The probability of more than one success occurring in such a short time interval or falling in such a small region is negligible.

The probability distribution of the Poisson random variable is called the Poisson distribution and is denoted by  $P(x; \mu)$  since its values depend only on  $\mu$ , the average number of successes occurring in the given time interval or specified region. This formula is given by the definition below:

**Definition:** The probability distribution of the Poisson random variable, representing the number of successes occurring in a given time interval or specified region is given by:

$$P(x; \mu) = \frac{e^{-\mu} \mu^x}{x!}; \quad x = 0, 1, 2, 3, \dots,$$

Where  $\mu$  is the average number of successes occurring in the given time interval or specified region and  $e = 2.71828$ .

**Theorem:** The mean and variance of the Poisson distribution  $P(x; \mu)$  both have the value  $\lambda$ .

#### Example 45

Suppose that an urn contains 100,000 marbles and 120 are red. If a random sample of 1000 is drawn what are the probabilities that 0, 1, 2, 3, and 4 respectively will be red.

$n = 1000$ ,

$$p = \frac{120}{100000}$$

$= 0.0012$ ,  $q = 0.9988$

#### Solution

Binomial =

$$\binom{1000}{x} 0.0012^x \times 0.9988^{1000-x}$$

For  $x=3$ ,

$$\binom{1000}{3} 0.0012^3 \times 0.9988^{997} = 0.0867$$

i.e  $166167000 \times 1.728^{-09} \times 0.30206$

Using the Poisson method,

$$\lambda = np = 1000 \times 0.0012 = 1.2$$

$$e - 1.2 = 0.3012$$

$$f(3) = \frac{e^{-1.2} 1.2^3}{3!} = 0.0867$$

$$\begin{aligned} P(X > 5) &= 1 - p(X \leq 4) \\ &= 1 - 0.9985 = 0.0015 \end{aligned}$$

#### Example 46

Let  $X$  have a Poisson distribution with a mean of  $\lambda = 5$ . Find

$$P(X < 6)$$

$$P(X > 5)$$

$$P(X = 6)$$

$$P(X > 4)$$

#### Solution

$$(i) P(X \leq 6) = \sum_{x=0}^6 \frac{5^x e^{-5}}{x!} = 0.762$$

$$(ii) P(X > 5) = 1 - P(X \leq 5) = 1 - 0.616 = 0.384$$

$$(iii) P(X = 6) = P(X \leq 6) - P(X \leq 5) = 0.762 - 0.616 = 0.146$$

$$(iv) P(X \geq 4) = 1 - P(X < 4)$$



**Example 47**

A hospital administrator, who has been studying daily emergency admissions over a period of several years, has come to the conclusion that they are distributed according to the Poisson law. Hospital records reveal that emergency admissions have averaged three per day during this period. If the administrator is correct in assuming a Poisson distribution. Find the probability that

- (i) exactly two emergency admissions will occur on a given day.
- (ii) No emergency admissions will occur on a particular day.
- (iii) Either 3 or 4 emergency cases will be admitted on a particular day.

**Solution**

(i)

$$p(X = 2) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\lambda = 3$$

=

$$\frac{e^{-3} 3^2}{2!} = \frac{0.05(9)}{2} = 0.225$$

$$(i) \quad p(X = 0) = \frac{e^{-3} 3^0}{0!} = 0.05$$

$$(ii) \quad p(X = 3) + p(X = 4) = \frac{e^{-3} 3^3}{3!} + \frac{e^{-3} 3^4}{4!}$$

=

$$e^{-3} \left[ \frac{27}{6} + \frac{81}{24} \right]$$

=

$$0.05 \left[ \frac{9}{2} + \frac{27}{8} \right]$$

$$= 0.05(7.875)$$

$$= 0.394$$

**Example 48**

Fit a Poisson distribution to the following data which gives the number of yeast cells per square for 400 squares.

| No.<br>of<br>cells<br>per<br>square<br>(x) | 0   | 1   | 2  | 3  | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Total |
|--------------------------------------------|-----|-----|----|----|---|---|---|---|---|---|----|-------|
| No.<br>of<br>squares<br>(f)                | 103 | 143 | 98 | 42 | 8 | 4 | 2 | 0 | 0 | 0 | 0  | 400   |

**Solution**

The expected theoretical frequency for  $r$  successes is  $Ne^{-m}m^r/r!$ , but  $m$  is not given in this example. The mean of the Poisson distribution is  $m$ . Hence

$$m = \frac{\sum fx}{\sum f} = \frac{529}{400} = 1.32$$

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{(1.32)^x e^{-1.32}}{x!}$$

thus,

$$f = 400 \frac{e^{-1.32} (1.32)^x}{x!}$$

, therefore,

| No.<br>of<br>cells<br>per<br>square<br>(x) | 0   | 1   | 2  | 3  | 4  | 5 | 6 | 7 | 8 | 9 | 10 | Total |
|--------------------------------------------|-----|-----|----|----|----|---|---|---|---|---|----|-------|
| No.<br>of<br>squares<br>(f)                | 107 | 141 | 93 | 41 | 14 | 4 | 0 | 0 | 0 | 0 | 0  | 400   |

#### 4.1.5 Relationship between the Binomial and Poisson distributions

In the binomial distribution, if  $n$  is large while the probability  $p$  of occurrence of an event is close to zero, so that  $q = 1 - p$  is close to one, the event is called a '**rare event**'. In practice, we shall consider an event as rare if the number of trials is at least 50 ( $n > 50$ ) while  $np$  is less than 5. For such cases the Poisson distribution very closely approximates the binomial distribution  $f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$  with  $\lambda = np$  (Spiegel, 1980). Thus the Poisson distribution is a limiting case of the Binomial distribution when  $n \rightarrow \infty$ ,  $p \rightarrow 0$  and  $np$  ( $n = \mu$ ) remain constant (Davis, 1986).

##### Example 49

In a manufacturing process in which glass is being produced, defects or bubbles occur, occasionally rendering the pieces undesirable for marketing. If it is known that on the average 1 in every 1000 of these items produced have one or more bubbles. What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles?

##### Solution

This is essentially a binomial experiment with  $n = 8000$  and  $p = 0.001$ . Since  $p$  is very close to zero and  $n$  is quite large, we shall approximate with the Poisson distribution, using  $\lambda = (8000)(0.001) = 8$ . Hence, if  $X$  represents the number of bubbles, we have:

$$\begin{aligned} P(X < 7) &= \sum_{x=0}^6 b(x; 8000, 0.001) \\ &= \sum_{x=0}^6 p(x; 8) \\ &= 0.3134 \end{aligned}$$

##### Post-Test

1. Suppose that 24% of a certain population have blood group B, for a sample of size 20 drawn

from this population, find the probability that

2. Exactly 3 persons with blood group B will be found.
3. Three or more persons  $\equiv$  the characteristics of interest will be found
4. Fewer than three will be found.
5. Exactly five will be found.
6. In a large population, 16% of the members are left-handed. In a random sample of size 10, find
7. The probability that exactly 2 will be left-handed  $p(X = 2)$

$$P(X \geq 2)$$

$$P(X < 2)$$

$$P(1 \leq X \leq 4)$$

1. Suppose mortality rate of a certain disease is 0.1, suppose 10 people in a community contract the disease, what is the probability that
2. None will survive
3. 50% will be
4. At least 3 will die
5. Exactly 3 will die
6. Suppose it is known that the probability of recovery from a certain disease is 0.4. If 15 people are stricken with the disease what is the probability that
7. or more will recover?
8. 4 or more will recover?
9. at least 5 will recover?
10. fewer than three recover?
11. In the study of a certain aquatic organism, a large number of samples were taken from a pond, and the number of organisms in each sample was counted. The average number of organisms per sample was found to be two. Assuming the number of organisms to be Poisson distributed. Find the probability that:
12. The next sample taken will contain one or more organisms.
13. The next sample taken will contain exactly three organisms.
14. The next sample taken will contain fewer than five organisms.
15. It has been observed that the number of particles emitted by a radioactive substance, which reach a given portion of space during time  $t$ , follows closely the Poisson distribution with parameter  $\lambda=100$ . Calculate the probability that:
16. No particles reach the portion of space under consideration during time  $t$ ;
17. Exactly 120 particles do so;
18. At least 50 particles do so.

19. The phone calls arriving at a given telephone exchange within one minute follow the Poisson distribution with parameter value equal to ten. What is the probability that in a given minute:
20. No calls arrive?
21. Exactly 10 calls arrive?
22. At least 10 calls arrive

### 4.3 Continuous Probability Distribution

#### 4.2.1 The Normal Distribution:

The graph of the normal distribution which is a bell-shaped smooth curve approximately describes many phenomena that occur in nature, industry and research. In addition, errors in scientific measurements are extremely well approximated by a normal distribution. Thus, the normal distribution is one of the most widely used probability distributions for modelling random experiments. It provides a good model for continuous random variables involving measurements such as time, heights/weights of persons, marks scored in an examination, amount of rainfall, growth rate and many other scientific measurements.

##### Definition:

The probability density function for the normal random variable  $X$  which is simply called normal distribution is defined by:

$$f(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, & -\infty < x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

Where  $\sigma > 0$ ,  $-\infty < x < \infty$  and the mean and variance of the measurements  $x$ , are  $E(x) = \mu$  and  $\text{Var}(x) = \sigma^2$ .

If the random variable is modelled by the normal distribution with mean,  $\mu$  and variance,  $\sigma^2$  then it is simply denoted as  $x \sim N(\mu, \sigma)$ .

This fact considerably simplifies the calculations of probabilities concerning normally distributed variables, as seen in the following illustration:

Suppose that  $X$  is  $N(\mu, \sigma^2)$ , let  $c_1 < c_2$ , and since  $P(X = c_1) = 0$ , then

$$\begin{aligned} P(c_1 < X < c_2) &= P(X < c_2) - P(X < c_1) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{c_2 - \mu}{\sigma}\right) - P\left(\frac{X - \mu}{\sigma} < \frac{c_1 - \mu}{\sigma}\right) \\ &= \Phi\left(\frac{c_2 - \mu}{\sigma}\right) - \Phi\left(\frac{c_1 - \mu}{\sigma}\right) \end{aligned}$$

Note that  $\Phi(-x) = 1 - \Phi(x)$ .

The normal distribution possesses the following properties.

##### Properties

Mean  $E(X) = \mu$

Variance  $\text{Var}(X) = \sigma^2$

Standard Deviation =  $\sigma$

### Example 50

1. If  $Z$  is  $N(0, 1)$ , find;
2.  $P(0.53 < Z < 2.06)$
3.  $P(Z > 2.89)$
4. If  $X$  is  $N(75, 100)$ , find
5.  $P(X < 60)$
6.  $P(6 \leq X \leq 12)$
7. If  $X$  is normally distributed with a mean of 6 and a variance 25, find  $P(6 \leq X \leq 12)$

### Solution

1. (i)  $P(0.53 < Z < 2.06) = \Phi(2.06) - \Phi(0.53) = 0.9803 - 0.7019 = 0.2784$
- (ii)  $P(Z > 2.89) = 1 - \Phi(2.89) = 1 - 0.9981 = 0.0019$
- (2)

$$P(X < 60) = P\left(\frac{X - 75}{10} < \frac{60 - 75}{10}\right) = P(Z < -1.5) = 0.0668$$

$$P(6 \leq X \leq 12) = P\left(\frac{6 - 6}{5} \leq Z \leq \frac{12 - 6}{5}\right) = P(0 \leq Z \leq 1.2)$$

$$= \Phi(1.2) - \Phi(0) = 0.8849 - 0.5000 = 0.3849$$

### 4.2.2 Normal approximation to the Binomial

Whenever the number of trials in a binomial experiment is small it is easy to find the probabilities associated with the experiment quickly as calculations become laborious. The normal distribution is often a good approximation to a discrete distribution when the latter takes on a symmetrical bell shape.

If  $X$  is a binomial random variable with mean  $\mu = np$  and variance  $\sigma^2 = npq$ , then the limiting form of the distribution of:

$$Z = \frac{X - np}{\sqrt{npq}} \quad \text{as } n \rightarrow \infty$$

is the standard normal distribution  $n(Z; 0, 1)$ .

It turns out that the normal distribution with  $\mu = np$  and  $\sigma^2 = npq = np(1 - p)$  not only provides a very accurate approximation to the binomial distribution when  $n$  is large and  $p$  is not extremely close to 0 or 1, but also provides a fairly good approximation even when  $n$  is small and  $p$  is reasonably close to.

In fact,

$$\frac{X - np}{\sqrt{npq}}$$

is asymptotically normal.

### Example 51

Consider a binomial distribution with p.d.f. of  $b(x; 15, 0.4)$ . Calculate the probability that  $x$  assumes values from 7 to 9 inclusive.

**Solution**

The exact probability is given by:

$$\begin{aligned}
 P(7 \leq X \leq 9) &= \sum_{x=7}^9 b(x; 15, 0.4) \\
 &= \sum_{x=0}^9 b(x; 15, 0.4) - \sum_{x=0}^6 b(x; 15, 0.4) \\
 &= 0.9662 - 0.6098 \text{ (from tables)} \\
 &= 0.3564
 \end{aligned}$$

Using normal approximation, corresponding values under a histogram:

$$\mu = np = 15 * 0.4 = 6$$

$$\sigma^2 = npq = 15 * 0.4 * 0.6 = 3.6$$

$$\therefore \sigma = \sqrt{3.6} = 1.897$$

(The 0.5 used here is called ‘Continuity Correction Factor’)  $z_1 = \frac{6.5-6}{1.897} = 0.26$

$$z_2 = \frac{9.5 - 6}{1.897} = 1.85$$

Now,

$$P(7 \leq x \leq 9) = p(0.26 < z < 1.85)$$

$$\begin{aligned}
 &= p(z < 1.85) - p(z < 0.26) \\
 &= 0.9678 - 0.6026 = 0.3652
 \end{aligned}$$

**Note:**

The 0.5 that was added to or subtracted from the 9 and 7 respectively is called “continuity correction factor”. It corrects for the fact that we are using a continuous distribution to approximate a discrete distribution. Generally, the degree of accuracy, which depends on how well the curve fits the histogram, will increase as  $n$  increases. If both  $np$  and  $nq$  are greater than or equal to 5, the approximation will be good.

Fig.4.4 shows the areas of the normal distribution representing probabilities within some defined standard deviations.

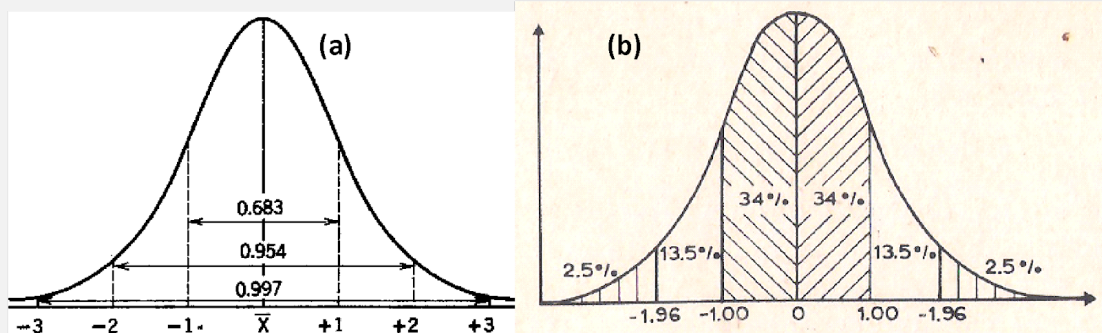


Fig. 4.4 Areas enclosed by successive standard deviation of the standard normal distribution.

### 4.2.3 Uniform Distribution

Suppose that a continuous random variable  $X$  can assume values in a bounded interval only, say the open interval  $(a, b)$ , and suppose the p.d.f. of  $X$  is given as

$$f(x; a, b) = f(x) = \frac{1}{b - a}, \quad a < x < b$$

$= 0$ , elsewhere.

This distribution is referred to as the Uniform or Rectangular Distribution on the interval  $(a, b)$  and is simply written as  $X \sim U(a, b)$ , where 'a' and 'b' are the parameters of the distribution. It provides a probability model for selecting a point at random from the interval  $(a, b)$ .

#### Properties

Mean

$$\mu = \frac{a + b}{2}$$

Variance

$$\sigma^2 = \frac{(b - a)^2}{12}$$

Standard Deviation  $\sigma =$

$$\sqrt{\frac{(b - a)^2}{12}}$$

#### Example 52

:

The hardness of a certain alloy (measured on Rockwell scale) is a random variable  $X$ . Assume that  $X \sim U[50, 75]$ .

- (a) Find  $P[60 < X < 70]$
- (b) Find  $E(X)$
- (c) Find  $\text{Var}(X)$

#### Solution

(a)

$$P[60 < X < 70]$$

$=$

$$\int_{60}^{70} \frac{1}{b - a} dx$$

$=$

$$\frac{1}{75 - 50} [x]_{60}^{70}$$

$=$

$$\frac{2}{5}$$

(a)  $E(X) =$

$$\frac{1}{b - a} \int_{50}^{75} x dx$$

$=$

$$\frac{125}{2}$$

Or

 $E(X) =$ 

$$\frac{b+a}{2} = \frac{75+50}{2} = \frac{125}{2}$$

(a)

$$\text{Var}(X) = E(X^2) - E^2(X) = \frac{1}{25} \int_{50}^{75} x^2 dx - \left(\frac{125}{2}\right)^2 = \frac{625}{12}$$

Or

$$\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{625}{12}$$

#### 4.2.4 The Gamma Distribution:

The Gamma distribution arises in the study of waiting times, for example, in the lifetime of devices. It is also useful in modeling many nonnegative continuous variables. The gamma distribution requires the knowledge of the gamma function.

**Definition:** The gamma function is defined by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

for

$$\alpha > 0$$

and for all  $x > 0$ 

$\Gamma(\alpha)$  is read as “gamma function of  $\alpha$ ”

Integrating by parts with  $u = x^{\alpha-1}$  and  $dv = e^{-x} dx$ , we obtain

$$\begin{aligned} \Gamma(\alpha) &= -e^{-x} x^{\alpha-1} \Big|_0^{\infty} + \int_0^{\infty} (\alpha-1) x^{\alpha-2} e^{-x} dx \\ &= (\alpha-1) \int_0^{\infty} x^{\alpha-2} e^{-x} dx \end{aligned}$$

Which yields the recursion formula

$$\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1).$$

Repeated application of the recursion formula gives

$$\begin{aligned} \Gamma(\alpha) &= (\alpha-1)(\alpha-2)\Gamma(\alpha-2) \\ &= (\alpha-1)(\alpha-2)(\alpha-3)\Gamma(\alpha-3) \end{aligned}$$

and so forth. Note that when  $\alpha = n$ , where  $n$  is a positive integer,

$$\Gamma(n) = (n-1)(n-2)(n-3) \dots \Gamma(1) = (n-1)!$$

#### General Gamma Distribution:

The continuous random variable has a gamma distribution, with parameters  $\alpha$  and  $\beta$  if its density function is given by:



$$f(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where  $\alpha > 0$  and  $\beta > 0$ .

#### Property of Gamma distribution

- (a)  $E(X) = \alpha\beta$
- (b)  $\text{Var}(X) = \alpha\beta^2$

#### 4.2.5 Exponential Distribution

An exponential distribution is a special gamma distribution for which  $\alpha = 1$ . It has many applications in the field of statistics, particularly in the areas of reliability theory and waiting times or queueing problems. It is a continuous distribution that can be related to the Poisson distribution in the discrete sense.

**Definition:** A continuous random variable has an exponential distribution with parameter  $\beta$  if its density function is given by:

$$f(x) = \begin{cases} \frac{1}{\beta} e^{-\frac{x}{\beta}} & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where  $\beta > 0$ .

#### Properties

Mean  $E(X) = \theta$

Variance  $\text{Var}(X) = \theta^2$

Standard Deviation  $\sigma = \theta$

So if  $\lambda$  is the mean of changes in the unit interval, then  $\theta = \frac{1}{\lambda}$  is the mean waiting time for the first change.

#### Example 53

Let the p.d.f. of  $X$  be  $f(x) = \left(\frac{1}{2}\right) e^{-x/2}$ ,  $0 \leq x < \infty$ .

- (i) What is the mean and variance of  $X$ ?
- (ii) Calculate  $P(X > 3)$
- (iii) Calculate  $P(X > 5 | X > 2)$
- (iv) Calculate  $P(X < 2)$

#### Solution

- (i)  $E(X) = \theta = 2$  and  $\text{Var}(X) = \theta^2 = 4$

- (ii)

$$P(X > 3) = \frac{1}{2} \int_3^\infty e^{-\frac{x}{2}} dx = e^{-\frac{3}{2}} = 0.2231$$

(iii)

$$P(X > 5 | X > 2) = \frac{\frac{1}{2} \int_5^{\infty} e^{-\frac{x}{2}}}{\frac{1}{2} \int_2^{\infty} e^{-\frac{x}{2}}} = \frac{e^{-\frac{5}{2}}}{e^{-\frac{2}{2}}} = e^{-\frac{3}{2}} = 0.2231$$

(iv)

$$P(X < 2) = \frac{1}{2} \int_0^2 e^{-\frac{x}{2}} = 1 - e^{-\frac{2}{2}} = 1 - e^{-1} = 0.6321$$

**NB.** Discuss other discrete and continuous distributions with students by stating their properties respectively.

### Post-Test

1. Let  $X$  have an exponential distribution with a mean of  $\theta = 20$ . Compute
2.  $P(10 < X < 30)$
3.  $P(0 < X < 30)$
4.  $P(X > 30)$
5.  $P(X > 40 | X > 10)$
6. Telephone calls enter a college switchboard according to a Poisson process on the

average of two every 3 minutes. Let  $X$  denote the waiting time until the first call that arrives after 10 A.M.

- (i) What is the p.d.f. of  $X$ ?
- (ii) Find  $P(X > 2)$
- (iii) Customers arrive randomly at a bank teller's window. Given that one customer arrived during a particular 10-minute period, let  $X$  equal the time within the 10 minutes that the customer arrived. If  $X$  is  $U(0, 10)$ , find
- (iv) The p.d.f. of  $X$ .
- (v)  $P(X \geq 8)$
- (vi)  $P(2 \leq X < 8)$
- (vii)  $E(X)$
- (viii)  $\text{Var}(X)$
- (ix) Explain the relationship that exists between the Poisson and the Exponential distributions.
- (x) If  $X$  is  $N(75, 100)$ , find  $P(X < 35)$  and  $P(70 < X < 100)$ .
- (xi) If  $Z$  is  $N(0, 1)$ , find values of  $c$  such that
- (xii)  $P(Z \geq c) = 0.025$
- (xiii)  $P(|Z| \leq c) = 0.95$
- (xiv)  $P(Z > c) = 0.05$
- (xv) Let  $X$  be  $N(\mu, \sigma^2)$ , so that  $P(X < 89) = 0.90$  and  $P(X < 94) = 0.95$ . Find  $\mu$  and  $\sigma^2$ .
- (xvi) Show that the random variable  $Z = (X - \mu)/\sigma$  is distributed  $N(0, 1)$ .
- (xvii) Suppose that  $Z \sim N(0, 1)$ . Find the following probabilities:
- (xviii)  $P(Z \leq 1.53)$

(xix)  $P(Z > -0.48)$

(xx)  $P(0.35 < Z < 2.01)$

(xxi)  $P(|Z| > 1.28)$

(xxii) Find the value of 'a' and 'b' such that

(xxiii)  $P(Z \leq a) = 0.648$

(xxiv)  $P(|Z| \leq b) = 0.95$